Supplementary Material for the paper "LUNAR: Automated Testing of Multi-Turn Conversations in Task-Oriented LLM Systems"

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Table 1: System intents and their descriptions.

Intent	Description
inform	The system provides information relevant to the user's query without a follow-up question or call to action.
inform_and_followup	The system provides information and additionally poses a follow-up question or proposes an action.
choice	The system offers at least two concrete options to choose from, with no follow-up question.
choice_and_followup	The system offers at least two options and additionally poses a follow-up question.
confirm	The system confirms the requested action, without a follow-up question.
confirm_and_followup	The system confirms the requested action and additionally poses a follow-up question.
clarify	The system asks the user a clarifying question to resolve ambiguity.
reject	The system indicates it cannot fulfill the request, without a follow-up question.
reject_and_followup	The system indicates it cannot fulfill the request and poses a follow-up question or suggests an alternative.
failure	The system signals that something went wrong (e.g. it did not understand the input).
misc	The system's response does not fit any other category.

Table 2: Intent transition mapping: overview of permitted user intents which can follow system intents.

System Intent	Permitted Follow-up User Intents
inform	add_preferences, ask, change_of_mind, stop
inform_and_followup	change_of_mind, add_preferences, confirmation, reject, reject_clarify, ask, stop
choice	choice, ask, add_preferences, change_of_mind, reject, reject_clarify, stop
choice_and_followup	choice, add_preferences, change_of_mind, reject, reject_clarify, stop
confirmation	change_of_mind, ask, add_preferences, stop
confirm_and_followup	change_of_mind, add_preferences, confirmation, stop
clarify	add_preferences, change_of_mind, ask, reject, stop
reject	change_of_mind, ask, stop
reject_and_followup	change_of_mind, confirmation, stop
failure	change_of_mind, repeat, stop
misc	change_of_mind, reject, reject_clarify, stop

Table 3: Effectiveness (mean failures found) split by SUT model averaged over 6 runs.

Approach	LLM	CarControl-II	CarControl-I	Navi-II	Navi-I
LUNAR	GPT-4o	13.29	159.33	120.67	419.83
	GPT-5-Chat	13.50	85.56	137.00	314.57
LUNAR_S	GPT-4o	8.83	78.50	73.83	125.00
	GPT-5-Chat	14.83	24.33	114.00	84.00
SENSEI	GPT-4o	0.14	23.57	31.38	23.50
	GPT-5-Chat	0.00	7.67	11.33	17.17

Table 4: Failure rate (mean critical ratio) split by SUT model average over 6 runs.

Approach	LLM	CarControl-II	CarControl-I	Navi-II	Navi-I
LUNAR	GPT-4o	0.099	0.470	0.371	0.591
	GPT-5-Chat	0.141	0.326	0.292	0.597
LUNAR_S	GPT-4o	0.071	0.233	0.278	0.256
	GPT-5-Chat	0.171	0.124	0.278	0.329
SENSEI	GPT-4o	0.001	0.103	0.109	0.051
	GPT-5-Chat	0.000	0.042	0.047	0.051

Table 5: Efficiency results (time per failure, hours/failure) split by SUT model. Lower is better.

Approach	LLM	CarControl-II(2.0h)	CarControl-I (2.0h)	Navi-II (3.5h)	Navi-I (4.0h)
LUNAR	GPT-4o	0.1505	0.0126	0.0290	0.0095
	GPT-5-Chat	0.1481	0.0234	0.0255	0.0127
LUNAR_S	GPT-4o	0.2264	0.0255	0.0474	0.0320
	GPT-5-Chat	0.1348	0.0822	0.0307	0.0476
SENSEI	GPT-4o	14.0000	0.0849	0.1116	0.1702
	GPT-5-Chat	–	0.2609	0.3088	0.2330